Universidade de Aveiro



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SynopStudio

AI Tool for Automatic Video Summarization and Editing

Milestone 3 - Digital Accessibility and Usability

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1. Legal Requirements

1.1. GDPR and Data Protection

1.1.1. Roles

- **Controller** for account data: email, display name, password hash, settings and logs.
- **Processor** for uploaded content: videos, transcripts and supporting materials. The user who uploads a video is responsible for having the right to use it, including the personal data of the people who appear in it. This is why the upload form requires a licence/permission declaration.

1.1.2. Personal data collected and purpose

Screen	Data	Purpose	Legal basis	Retention
Create account	Email	Authentication, account notices, abuse handling	Contract	While the account exists
Create account	Password	Authentication. Credentials are managed by Keycloak (identity provider) and stored only as a salted hash using the algorithm configured in the realm's password policy	Contract	While the account exists
Create account	Display Name	Shown in the interface only	Contract	While the account exists

Create account	Age declaration	Compliance with the Terms (minimum age 13)	Contract	While the account exists
New Project / Summarize	Video file or YouTube link (faces and voices of third parties)	Generating clips or summaries	Contract	30 days after last access (proposed)
New Project / Summarize	Supporting materials (PDF, PPTX, DOCX, TXT)	Improving transcription accuracy	Contract	Same as the video
New Project / Summarize	Title, description, focus text	Context for the AI	Contract	Same as the video
Derived	Transcripts, speaker segments, generated clips	Editing and traceability	Contract	Same as the video
Settings	Notification preferences	Email notifications	Contract	While the account exists
Implicit	IP address, session cookies, activity and AI-decision logs	Security, abuse prevention, traceability	Legitimate interest	12 months, pseudonymised

1.1.3. Anonymisation and pseudonymisation

- **Video content cannot be realistically anonymised**, because faces and voices remain identifiable. It is protected through access control and encryption instead.
- **Logs** store a hashed user ID instead of the email address.
- **Speaker diarisation** produces a voice profile (Speaker.voiceProfile) for each speaker, scoped to a single video. Profiles are used only to distinguish speakers within that video, are never compared across videos or users, and are permanently deleted together with the project. Because they are not used to uniquely identify a person, they are not treated as biometric data under Art. 9; however, they are protected with the same encryption and access controls as the video itself.
- **Special categories:** transcripts of political content may reveal political opinions (Art. 9). They are not used for any purpose other than the user's own project.

1.1.4. Third parties

Third parties receiving data:

Recipient	Data Shared	Notes
Hosting provider	All stored data	EU region
Email provider	Email address, notification content	Clips ready, Weekly digest
Invited users	Project content (video, transcript, clips)	Only for projects the owner explicitly shares

1.1.5. Data Subject Rights

Users can permanently delete a single project or their entire account from the application. Archiving a project does not delete it.

- **Project deletion** removes the original video, supporting materials, transcripts, speaker data, generated clips and cached files, and revokes access for invited users.
- **Account deletion** deletes all the user's projects and removes the account from Keycloak.

Data is removed from active storage immediately and from backups within 30 days. Requests are answered within one month, as required by Art. 12(3). Pseudonymised security logs are kept for 12 months based on legitimate interest (Art. 6(1)(f)), as listed in section 1.1.2. Audit records of licence confirmations and editorial decisions are also kept for 12 months, as they may be needed for the establishment, exercise or defence of legal claims (Art. 17(3)(e)). Both are deleted after that period. Full conditions are described in the Terms and Conditions (Section 1.4.3).

Users can also correct their name and email in Settings (rectification, Art. 16) and download their transcripts and clips (portability, Art. 20).

1.1.6. Security Measures

- **Encryption in transit:** all connections use TLS 1.3.
- **Encryption at rest:** videos, documents and transcripts are encrypted with AES-256, as defined in the non-functional requirements (MS2).
- **Authentication:** handled by Keycloak, including password policy (minimum 8 characters) and brute-force detection.
- **Access control:** a project and its media can only be accessed by its owner and by users the owner explicitly invites. Media files are served through signed, time-limited URLs.

1.2. Licensing and Intellectual Property

The SynopStudio software stack is built on open-source tools and models, and we checked each one against this requirement. We also audited the license of every third-party component in the MS2 architecture, along with the AI models and media libraries the workers use. For each component we took the license from the official repository or model card. We then checked what each license requires of SynopStudio, given how the component is used: run as a separate service, linked as a library or loaded as model weights.

Component	License	Usage
React	MIT	Library
FastAPI	MIT	Library
Traefik	MIT	Separate Service
Keycloak	Apache-2.0	Separate Service
RabbitMQ	MPL-2.0	Separate Service
PostgreSQL	PostgreSQL License	Separate Service
SeaweedFS	Apache-2.0	Separate Service

faster-whisper + CTranslate2	MIT	Library
pyannote.audio	MIT	Library
pyannote speaker-diarization-community-1	CC-BY-4.0	Model Weights
FFmpeg	LGPL-2.1 (default build)	External Process
yt-dlp	Unlicense	Library
pypdf	BSD-3	Library
python-pptx	MIT	Library
python-docx	MIT	Library
Qwen3.8-27B	Apache-2.0	Model Weights

After reviewing the license of every component and finding no conflicts between them, the team chose to release SynopStudio under Apache-2.0. Most of the stack already uses permissive licenses (MIT, Apache-2.0, BSD). Also Apache-2.0 allows SynopStudio to be adopted by an institution or turned into a commercial product later.

As mentioned in MS2, much of SynopStudio's stack and design is based on SupoClip, an open-source alternative to OpusClip. SupoClip is released under AGPL-3.0, a highly restrictive license: any project that copies or adapts its code must also be released under AGPL-3.0. SynopStudio is a web application, so its full source would have to be offered to every user.

The team uses SupoClip only as a reference for stack and design decisions and does not copy any of its code. Copyright does not cover ideas or architectural choices, so this use creates no AGPL-3.0 obligations, and SynopStudio can be released under Apache-2.0.

1.3. Sector-specific regulations

SynopStudio has no single sector regulator. Health has the ERS and payments have PSD2, but a tool that turns long recordings into short clips falls under three sectors at once. Each has its own rules:

1. Artificial intelligence, because the platform transcribes, selects clips and generates text with AI models.
2. Audiovisual media and copyright, because it handles third-party recordings and the images and voices of real people.
3. Political and institutional communication, because its main use cases are academic lectures and political debates.

1.3.1. Artificial intelligence and AI-generated content

The EU AI Act (Regulation (EU) 2024/1689) is the main rule for this sector.

Risk classification

SynopStudio is not a high-risk system. The high-risk list (Annex III) includes AI intended to influence elections or voting behavior. SynopStudio summarizes recordings faithfully and treats social media engagement only as a secondary indicator. It is therefore a limited-risk system, subject to the transparency obligations of Article 50, which apply since 2 August 2026.

Transparency obligations (Art. 50)

Providers must mark AI-generated text, audio, and video in a machine-readable way so it can be detected as artificial. For systems already on the market before 2 August 2026, the Digital Omnibus on AI (Regulation (EU) 2026/1744) gives until 2 December 2026. Whoever publishes AI-generated text on matters of public interest must disclose it, unless the text was reviewed by a person who takes editorial responsibility. Deepfakes must always be disclosed.

In SynopStudio:

- AI-generated titles, descriptions and keywords are labelled as AI text (FR16).
- No clip is exported without the creator's explicit approval (FR15). Together with the audit log of AI decisions and user changes (FR18), this provides the human review that the Act requires for public-interest text.
- Low-confidence transcripts and generated text are highlighted for review (FR13).
- The platform only cuts and reframes real footage. It never synthesizes voices or faces, so it does not produce deepfakes.
- Machine-readable marking of the exported AI text is obligatory. C2PA is being evaluated for it.

1.3.2. Audiovisual media and copyright

Every project starts from a recording that someone owns or is authorized to use. That recording shows identifiable people, so copyright, image rights and the rules of video platforms all apply.

Copyright (CDADC)

The Portuguese Copyright Code (CDADC), updated in 2023 to transpose the EU Digital Single Market Directive (Decree-Law 47/2023), treats our two main content types differently:

- Political speeches: speeches in assemblies, councils, and public debates are not protected (Art. 7). Third-party use must still be limited to what its purpose requires.
- Lectures: professors' lectures may only be published by third parties with the author's authorization (Art. 79).
- Quotations: short extracts may be used for criticism, discussion or teaching, provided the author and source are identified (Arts. 75 and 76).
- Moral rights: authors keep the right to oppose changes that distort their work or harm their reputation (Art. 56).

Image, voice and minors

- Image and voice are personal data.
- Under the Civil Code (Art. 79), a portrait cannot be shown or reproduced without consent. The exceptions are notoriety or public office, scientific, teaching or cultural purposes, and images taken in public places or public events. Even then, it cannot be used in a way that harms the person's honor or reputation.
- For minors, Portugal set the age of digital consent at 13 under the GDPR. Below that age, consent comes from the holders of parental responsibility.

Video platforms

- YouTube’s Terms of Service only allow content to be downloaded where the service expressly permits it, or with prior written permission from YouTube and, where applicable, the rights holders.
- YouTube also requires creators to disclose realistic, altered, or synthetic content.
- Because SynopStudio stores videos and shares clips through links, it is a hosting service under the Digital Services Act (Regulation (EU) 2022/2065). In Portugal this is

supervised by ANACOM under Law 12-A/2026. Every hosting service must offer a way to report illegal content and must give reasons when it removes content.

In SynopStudio

- The AI pipeline only starts after the creator confirms authorship and selects a license (FR01). A lecture uploaded by someone other than the lecturer requires the lecturer's authorization.
- Each clip keeps a verifiable link to the exact moment in the original recording, which identifies the source (FR11).
- Context-loss warnings and the ban on fabricated quotes (FR11, FR15) protect speakers' reputation and the integrity of their work.
- Clips can be shared with anyone who has the link, and projects can be fully deleted (UC20, UC21).
- Link import (FR02) is only available for videos the creator owns or is authorized to use. Under SynopStudio's Terms of Service, the creator is responsible for holding that right, and the authorization is confirmed together with the license in FR01. Direct upload remains the default.
- The Viewer page will include a "report this clip" option, handled by the Administrator's moderation rules (UC24).

1.3.3. Political and institutional communication

Councilors, lecturers, and student associations use SynopStudio to communicate with the public. Clips of political debates must therefore follow political advertising and electoral rules and must not distort what was said.

Political advertising (Regulation (EU) 2024/900)

This regulation has fully applied since 10 October 2025. Paid political advertisements must be labelled with who paid for them, how much they cost and, where applicable, the audience targeted.

Electoral periods (Law 72-A/2015, Art. 10)

Once the decree setting an election date is published, commercial advertising cannot be used for political propaganda in any medium, including social networks.

Editorial neutrality

The project proposal requires rules on editorial neutrality. A short clip taken out of context can change the meaning of a political or scientific statement.

In SynopStudio:

- SynopStudio does not sell or place advertisements. However, a creator who pays to promote a political clip turns it into a political advertisement. The export screen will warn about this and about the ban during electoral periods.
- Every proposed clip comes with the reason it was selected, and every clip links back to the exact moment in the original recording (FR10, FR11). Authors of the project can check the full context, which matches the motivation of the councilor persona. Viewers can check only the final edited clip.
- AI-generated text never attributes to a speaker words they did not say (FR15), and all automatic decisions and user edits are logged (FR18).

1.3.4. Technical standards and non-applicable rules

Adopted standards

- W3C WebVTT: the web format for captions and subtitles. It is used for the independent subtitle files in the export (FR19).
- C2PA / Content Credentials: an open provenance standard that records origin, edits and AI use in a signed manifest. It is being evaluated as the machine-readable marking required by Article 50 of the AI Act.
- OpenAPI 3.x: used to document the API, as required by the non-functional requirements.
- ISO/IEC 42001 (AI management systems): used only as a reference for governance practices such as human approval and audit logging.

Judged not applicable

- ISO 20000 / ITIL: these IT service management frameworks do not apply, because SynopStudio is an academic prototype and not an operated IT service.
- Audiovisual Media Services Directive: SynopStudio is not a video-sharing platform. It is a production tool, and clips are shared through individual links, not through a public catalogue of videos. This would need to be reviewed if a public gallery were added.

1.4. Terms and Conditions

1.4.1. Acceptable Use Policy

Responsibilities

Users must be at least 13 years old to use SynopStudio's services. However, minors between the ages of 13 and 18 may use the services if authorized by a parent or legal guardian.

Users must provide accurate registration information, protect their login credentials, and are responsible for all account activities.

Permissions

- Create projects with clearly stated licenses, copyrights, and contributors, confirming that the user owns the rights or has authorization to use them;
- Include supporting files for which the user owns the rights or has authorization to use and modify them;
- Upload videos either directly or via links from legal platforms that allow it (e.g., YouTube, institutional repositories) in line with their terms of service;
- Process duly authorized lectures, classes, debates, conferences, and meetings.

Restrictions

- Submitting videos and supporting documents containing illegal content or content that infringes on the rights of third parties;
- Submitting videos containing recordings of people without their consent or authorization, or content featuring minors without the authorization of their legal guardians;
- Using text-based editing to cut, reorder, or combine speech in a way that alters the original meaning, whether through image manipulation, creating disinformation videos, or taking speech out of context;
- Submitting defamatory, discriminatory, sexually explicit, or violent content;
- Removing or hiding indications that content was generated by AI and links to the original content;
- Using the service in an automated manner that compromises its availability;
- Sharing accounts or circumventing license verification.

Consequences and Complaints

Any violation may result in the removal of content or the suspension or permanent banning of the account, with notification of the reason, the option to export data, and the opportunity to appeal against the decision. The decision is saved and recorded in the log.

Anyone can report content that infringes rights or alters the original meaning, and the content may be suspended for pending review.

1.4.2. Liability Disclaimers and AI

The software is provided "as is," without any express or implied warranties of uninterrupted operation or against losses resulting from system failures.

SynopStudio assumes no legal liability to the extent permitted by law and does not guarantee absolute accuracy in the results due to inaccuracies in subtitles or transcripts, even when supported by supplementary documents provided by the user; errors in distinguishing or attributing lines to specific speakers when multiple participants are involved; or any semantic inaccuracies in titles, summaries, or suggestions.

The platform is not liable for data loss, delays in processing queues, or the temporary unavailability of storage services and external APIs (e.g., YouTube).

The user is solely responsible for reviewing, correcting, and explicitly approving the text, cut points, speaker attribution, and framing before exporting. The user is responsible for the content they publish and agrees to be liable for third-party claims, defamation lawsuits, privacy violations, or copyright infringements.

1.4.3. Refund/Cancellation Policies

In its current version, SynopStudio is provided free of charge, without paid subscriptions, token purchases, or billing data collection.

Users may choose to delete specific projects or permanently close their account at any time, which results in the permanent deletion of uploaded videos, auxiliary files, transcripts, and generated and exported clips from the platform servers. The only records that remain are the required billing records and audit records related to the confirmation of licenses and editorial decisions, which are retained for a period of 12 months for legal compliance purposes.

SynopStudio may introduce paid monthly or annual subscription plans and processing tokens in future commercial versions. If implemented, users will be able to cancel their subscription at any time before the automatic renewal date. After requesting cancellation, the user retains access to all paid features until the end of the billing period already paid for, after which unused monthly tokens expire, and the account reverts to the free plan at no additional cost.

Under EU consumer law (Directive 2011/83/EU and Portuguese Decree-Law 24/2014), users will be able to withdraw from a future paid plan within 14 days of purchase without giving a reason. If the user expressly requests to start using the plan immediately upon purchase, the refund will be reduced in proportion to the tokens already used (for example, imports, transcriptions, or clip generations). If all tokens of the plan have been used, the right of withdrawal no longer applies, as informed and agreed at purchase.

1.4.4. Legal Jurisdiction

These terms are governed by Portuguese law, without prejudice to the mandatory European Union regulations applicable to the service, specifically regarding data protection, artificial intelligence, digital services, and copyright (see Section 1.3) and to the consumer protection provisions of the law of the user's country of residence, when such provisions are more favorable to the user.

1.4.5. Changes to the Terms

The terms may be amended to reflect changes in the service, the law, or security requirements. Each version is numbered and dated, and previous versions remain available for reference.

Material changes to the terms and conditions (including future introduction of paid subscription plans) are communicated two months in advance, via email and on the web application, indicating the change and the corresponding decision. Non-material changes (corrections and clarifications) are published with the new date.

Changes to privacy policies that alter the purposes or categories of data processed are communicated to the user before they take effect.

Changes required by law or for urgent security reasons may take effect sooner, with notice provided as soon as possible.

Changes to the service (new features or discontinuation) follow the same principle: advance notice when they negatively affect usage, with the right to cancel at no cost.

The terms are reviewed at least once a year.

2. Usability and Accessibility

2.1. Accessibility Standards and User Profiles

2.1.1. Accessibility Standards

WCAG 2.2, Level AA. SynopStudio adopts WCAG 2.2 Level AA as its conformance target for all screens of the web application. This version includes criteria that are directly relevant to a video-editing tool, namely:

- 2.5.7 Dragging Movements (clip trimming on the timeline);
- 2.5.8 Target Size (word-level selection in the transcript);
- 2.4.11 Focus Not Obscured (sticky video player over the transcript).

Decree-Law 83/2018 (Portugal). This law transposes Directive (EU) 2016/2102 and requires public sector websites and mobile applications to conform to accessibility standards. Since the project is developed at the University of Aveiro, a public body, it would apply if the tool were deployed by the University. The WCAG 2.2 AA target exceeds the level referenced by this law.

European Accessibility Act (Directive (EU) 2019/882). The EAA applies to specific services such as e-commerce, banking, transport and electronic communications. A video summarisation and editing tool does not fall within these categories, so the EAA is not directly applicable. The project nevertheless follows equivalent principles.

2.1.2. User Profile Framework

We review every SynopStudio screen against six user profiles, each tied to an MS2 persona or actor. The clip editor (timeline with Text, Overlay, Video and Audio tracks) is the hardest screen for almost every profile: it relies on dragging, icon-only buttons and small text on a dark background.

Profile	SynopStudio scenario	Main barriers	Test question	Key WCAG criteria
Blind Users	A Viewer opens a shared link and plays the clip; A creator creates a project.	Unlabelled player controls; link to the original shown only on the timeline; progress not announced	Can I find and play a shared clip? (Viewer) Create a project, using only a screen reader? (Creator)	1.1.1, 4.1.2, 4.1.3
Low vision	Eduardo reviews a transcript at 200% zoom	Low-contrast grey text; editor layout breaks when zoomed	Is everything readable at 200% zoom?	1.4.3, 1.4.4, 1.4.10
Color Blindness	João reviews many clips a week	Speakers, AI confidence and clip status shown only by color	Can I understand everything without color?	1.4.1, 1.4.11
Motor impairments	Luciano edits on his laptop using only the keyboard	Small trim handles; drag-only reordering and upload	Can I cut and approve a clip with the keyboard only?	2.1.1, 2.5.7, 2.5.8
Deafness	A deaf creator must confirm captions match the audio (FR15)	Review assumes hearing the audio; sound-only alerts	Can I review and approve a clip without sound?	1.2.2, 4.1.3
Cognitive / low digital literacy	António and Paula, with no editing experience	Jargon (codec, aspect ratio, CC licences); long flow; unclear errors	Do I know what to do on the first try?	3.3.1, 3.3.2, 3.3.3

2.2. Screen Reader and Semantics

SynopStudio has to work with screen readers, the software that blind and low vision users rely on to read the interface aloud. A screen reader does not see the screen. It reads the structure of the page and announces the role, name and state of each element while the user moves through the page with the keyboard. For this to work, the structure of each page has to be correct and its semantics have to match what each element does: native HTML elements used for this purpose, and ARIA attributes only where the HTML is not enough, as in the custom player and timeline. This section covers the screens involved, page structure and navigation, the name and the states of controls.

2.2.1. Screens for screen readers

Screen	Use Case	Elements a screen reader must announce
Landing Page	---	Headings, links to sign-in and sign-up
Sign-in and sign-up	---	Field labels, password rule, error messages
Project list	---	Search fields, filters, project cards and their status, result count
New Project	UC01, UC02, UC03	File input and accepted formats, title, description, upload progress
Summarize	---	Generation settings, processing progress
Clip proposals	UC10, UC11	Player controls, clip list and selected clip, source timeline
Timeline Editor	UC12, UC13	Three panels and the timeline, tracks with visibility and sound toggles, clip edges, tool buttons
Share Dialog	UC20	Share link, "Copy" button and its confirmation, invited people

Settings	---	Profile fields, "Save Changes", "Log out" and "Delete Account" buttons, notification switches and their state
Viewer page	UC25	Player, transcript, link to the original moment

2.2.2. Page Structure and Navigation

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
1.3.1 Info and Relationships	A	Information, structure and relationships shown visually can be determined in code	Each page has one title as its <h1> and its sections as <h2>. The sidebar is a <nav>, the main area a <main>, and the project and clip lists are marked up as lists.
1.3.2 Meaningful Sequence	A	When the order of content affects its meaning, the correct reading order can be determined in code	The timeline editor has three panels and a timeline. Their order in the code follows the order of the work, so the screen reader reads them in the sequence the user edits, not in the order they are laid out on screen.
2.4.2 Page Titled	A	Each page has a title that describes its topic or purpose	The browser title changes on every page, for example "Projects SynopStudio" or the project's name.
2.4.6 Headings and Labels	AA	Headings and labels describe their topic or purpose	Headings name the content below them (for example, the heading above the clip list), so a user who jumps between headings knows where they are.
3.1.1 Language of Page	A	The default language of each page can be determined in code	The interface is in English and declares lang="en", so the screen reader uses an English voice.
3.1.2 Language of Parts	AA	The language of each passage or phrase in another language can be determined in code	The "Português (PT)" and "Español (ES)" buttons, and transcript and titles in Portuguese, declare their own lang attribute (lang="pt-PT", lang="es"), so the screen reader does not read them with an English voice.

2.4.4 Link Purpose	A	The purpose of each link is clear from its text or its context	Links such as the back link to the project list and the sign-up link describe where they lead. Arrows and other decorative characters are hidden from screen readers.

2.2.3. Names, Roles and State of Controls

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
1.1.1 Non-text Content	A	Non-text content has a text alternative; decorative content is ignored by assistive technology	Waveform illustrations on project cards and on the sign-in page are decorative and hidden. The waveform timeline carries information (where each clip sits), so its text alternative is the clip list with start and end times.
4.1.2 Name, Role, Value	A	Each control exposes its name and role, and its state can be read and set by assistive technology	Every icon-only button has an accessible name: play and log out, and in the timeline editor cut, duplicate, delete, undo, redo, zoom in, zoom out, snap, and the visibility and sound toggles on each track. The track toggles, the snap tool and the notification switches expose their on/off state. The selected clip exposes that it is selected, and the timeline is a slider whose value is read as a time.

2.2.4. Dynamic Content and Status Messages

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
4.1.3 Status Messages	AA	Status messages can be presented by assistive technology without receiving focus	Upload and processing progress, the "Copy" confirmation in the share dialog, the "Save Changes" confirmation, the number of search results and the clip now playing are announced through live regions without moving focus.

2.3. Color, Contrast and Legibility

SynopStudio uses a light theme for the app screens and a dark theme for the clip editor, with black primary buttons and an orange accent (#E54823). Every color token meets WCAG 2.2

AA in both themes: 4.5:1 for text and 3:1 for borders, icons and controls. The orange accent is used where 3:1 is enough (selection, toggles, focus ring, large headings); a deeper orange (#C93C1A) is used for small text and the Export button.

2.3.1. Color palette

Token	Light (app screens)	Dark (clip editor)	Required	Contrast light	Contrast dark
Background	**#FFFFFF**	**#111111**	—	—	—
Surface	**#F6F6F6**	**#1D1D1D**	—	—	—
Divider	**#E2E2E2**	**#383838**	—	—	—
Border	**#858585**	**#858585**	3:1	3.4:1	4.6:1
Text	**#111111**	**#FFFFFF**	4.5:1	17.5:1	16.9:1
Text secondary	**#707070**	**#9F9F9F**	4.5:1	4.6:1	6.4:1
Text muted	—	**#8A8A8A**	4.5:1	—	4.9:1
Icons	**#707070**	**#8A8A8A**	3:1	4.6:1	4.9:1
Primary button	**#111111**	**#FFFFFF**	4.5:1	18.9:1	18.9:1
Accent	**#E54823**	**#E54823**	3:1	3.7:1	4.2:1
Accent text	**#C93C1A**	**#F26B4B**	4.5:1	4.7:1	5.6:1
Accent button	**#C93C1A**	**#C93C1A**	4.5:1	5.1:1	5.1:1
Focus ring	**#E54823**	**#E54823**	3:1	3.7:1	4.2:1
Success	**#15803D**	**#4ADE80**	4.5:1	4.6:1	9.7:1
Warning	**#854D0E**	**#FACC15**	4.5:1	6.3:1	11.0:1
Error	**#B91C1C**	**#F87171**	4.5:1	6.0:1	6.1:1

AI label	**#6D28D9**	**#C4B5FD**	4.5:1	6.6:1	9.1:1

2.3.2. Contrast and legibility criteria

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
1.4.1 Use of Color	A	Color is not the only way to convey information	Project status is written as text (Draft, Archived). Selected Format and Style cards show a check mark, and toggles show an on/off label. In color-blindness simulations the orange accent and the success green look similar, so every status has an icon and text.
1.4.3 Contrast (Minimum)	AA	Text has 4.5:1, large text 3:1	All text tokens reach at least 4.5:1 (table 2.3.1). Small orange text and the Export button use #C93C1A (5.1:1 with white text). The orange heading on the landing page is large text and uses #E54823 (3.7:1).
1.4.11 Non-text Contrast	AA	Controls, borders and icons have 3:1	Input and search borders use #858585 (3.4:1). Editor icons use #8A8A8A (4.9:1). Orange selection borders and the focus ring reach 3.7:1.
1.4.4 Resize Text	AA	Text can be zoomed to 200% without loss	Sizes use rem, and labels such as the durations on project cards grow with their text, so nothing is cut off at 200%.
1.4.10 Reflow	AA	At 320 px wide, no scrolling in two directions	The sidebar becomes a menu and the Format and Style cards stack. The editor timeline is exempt, because it needs a two-dimensional layout, but its panels stack.
1.4.12 Text Spacing	AA	More line, word and letter spacing does not hide text	Buttons and cards have no fixed heights, so long titles such as "Open Lecture — Distributed Systems" wrap.

Subtitles. The Classic, Bold and Minimal styles draw white text with a black outline, and Boxed uses its box, so subtitles stay readable on any video frame. Any new color is added to the palette and checked in both themes before use.

Tools: WebAIM Contrast Checker and Chrome DevTools (Rendering → Emulate vision deficiencies).

2.4. Keyboard Navigation, Focus and Form Interaction Audit

SynopStudio should allow users to complete the entire workflow, from creating an account and uploading a video to editing and exporting clips, using only the keyboard.

This section evaluates keyboard operation, focus management, button sizes and other features, and form interactions across the application's various screens, with the goal of overcoming the barriers faced by different user profiles, including those with motor and cognitive limitations or low digital literacy.

2.4.1. Keyboard and Focus

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
1.4.13 Content on Hover or Focus	AA	Additional content that appears on pointer hover or keyboard focus is dismissible, hoverable and persistent until focus or hover is removed.	Tooltips showing AI clip selection rationale (FR10, US17), context-loss warnings (FR11, US19), and low-confidence word suggestions in the transcript (FR13, US22) must stay visible when hovered and close when pressing "Esc".
2.1.1 Keyboard	A	All functionalities can be operated using only the keyboard, without requiring specific timing of keystrokes.	Deleting or highlighting text, setting start and end points, and reordering or rejecting segments (US20, US21) must work without a mouse, as must approve a clip (US24).
2.1.2 No Keyboard Trap	A	If keyboard focus can enter a component, it can also leave it using only the keyboard.	Several components capture focus: the player and timeline in the editor (FR12), the preview players in clip proposals (FR10), and modal dialogs such as project deletion (US34) and clip approval (FR15).
2.1.4 Character Key Shortcuts	A	Single-character shortcuts can be turned off, remapped, or only	The editor may offer single-key shortcuts such as play/pause or split (FR12). They must be

		work when the relevant component has focus.	configurable or active only while the editor has focus.
2.2.1 Timing Adjustable	A	Any time limit can be turned off, adjusted or extended by the user, unless it is essential or part of a real-time event.	The session and the project forms must not expire while the user waits for processing or fills in a long form.
2.4.1 Bypass Blocks	A	A mechanism allows users to skip blocks of content repeated across pages.	All screens share the same navigation area. In the editor, a keyboard user needs a way to jump to the main content instead of tabbing through the navigation every time.
2.4.3 Focus Order	A	Focusable components receive focus in an order that preserves meaning and operability	Tab order must follow the editing flow and stay logical when panels open, such as the clip review with source intervals and context-loss warning (FR11).
2.4.7 Focus Visible	AA	Any keyboard-operable interface has a visible focus indicator.	Each focusable element needs a visible indicator, so the user always knows which word, segment or clip is selected.
2.4.11 Focus Not Obscured	AA	A component that receives focus is not entirely hidden by content created by the author.	Fixed elements such as the player, the top bar and the processing progress notifications (FR02) can cover the focused element while the user scrolls through a long transcript.
2.5.2 Pointer Cancellation	A	For actions operated using a single pointer, activation occurs on the up-event (release) and can be aborted by moving the pointer away before releasing or undone after completion.	Deleting a project (US34), rejecting a clip segment (US21), or releasing a timeline trim handle (FR12) must trigger only on mouse/touch release and allow cancellation by dragging away.

2.5.3 Label in Name	A	For components with a visible text label, the accessible name contains that visible text.	Export, generate clips or reject segment (FR09, FR15, FR19), must expose an accessible name that contains the same text.
2.5.7 Dragging Movements	AA	Any action that uses dragging can also be done with a single pointer without dragging, unless dragging is essential.	Trim handles, segment reordering on the timeline (US21) and drag-and-drop upload of videos and context documents (FR02, FR03) need a non-drag alternative.
2.5.8 Target Size (Minimum)	AA	Pointer targets are at least 24x24 CSS pixels, unless spacing or another exception applies.	Trim handles on the timeline, word-level elements in the transcript (FR04) and icon buttons on player controls and clip cards are likely to be below 24x24 px.

2.4.2. Form Interaction Audit

WCAG 2.2 Criteria	**Level**	**Requirement**	**SynopStudio**
1.3.1 Info and Relationships	A	Information, structure, and relationships conveyed through presentation can be programmatically determined or are available in text.	Form labels must be explicitly linked to inputs, related options in clip configuration (FR09) must be grouped with <fieldset> and <legend>, and validation errors or format hints must be bound via aria-describedby.
1.3.5 Identify Input Purpose	AA	Input fields that collect information about the user have a programmatically identified purpose (ex: Autocomplete).	Account creation collects email, password and display name (section 1.1.2), and login repeats email and password. These fields need a programmatic input purpose so browsers and assistive tools can fill them in.
2.4.6 Headings and Labels	AA	Headings and labels describe topic or purpose of the content.	The project form, the video link and time range need labels that say what to enter, in plain language rather than terms such as codec or aspect ratio.

3.2.1 On Focus	A	Moving focus to a component does not trigger an unexpected change of context	Focusing on a field must not start an action (ex: focusing on the video link or the time range field must not start validation, import or navigation).
3.2.2 On Input	A	Changing a setting or entering data does not trigger an unexpected change of context, unless the user is warned beforehand.	Selecting a file or pasting YouTube link (FR02) or changing the duration or format of a clip (FR09), must not start processing or change screen without warning. Import and generation start only from an explicit button, after license confirmation (FR01).
3.2.3 Consistent Navigation	AA	Navigation mechanisms repeated across pages appear in the same relative order.	The sidebar and the navbar must appear in the same place and order on every screen of the flow.
3.2.4 Consistent Identification	AA	Components with the same function are identified consistently across pages.	Actions with the same function keep the same name and icon on every screen. AI-generated text is always labelled in the same way (US27).
3.2.6 Consistent Help	A	Help mechanisms that are repeated across multiple web pages occur in the same relative order to the other page content.	Help links, documentation on license types (FR01), supported media/document formats (FR02, FR03) and keyboard shortcut guides in the editor must appear in the same consistent location header or form help across all screens.
3.3.1 Error Identification	A	When an input error is detected automatically, the field in error is identified and the error is described in text.	The system rejects videos without audio, corrupted files and invalid links (FR02, US07) and blocks processing without licence confirmation (FR01).
3.3.2 Labels or Instructions	A	Labels or instructions are provided when content requires user input.	Required fields and formats must be shown before submitting (licence and authorship declaration (FR01), accepted video formats (MP4, MOV, WebM, MKV, AVI) (FR02), document formats (FR03) and the

			start and end time format (US05)).
3.3.3 Error Suggestion	AA	When an error is automatically detected and suggestions for correction are known	Each message should say how to use a supported format, enter a public or authorised link, or choose a time range inside the video duration.
3.3.4 Error Prevention	AA	Submissions that create legal commitments or modify or delete user data are reversible, checked, or confirmed before being finalized.	Delete account and deleting project and all its data (US34), approving a clip for export (FR15, US24) and confirming the licence and authorship declaration (FR01) need review or confirmation.
3.3.7 Redundant Entry	A	Information already entered in the same process is auto populated or available to select, instead of being asked again.	Authorship, speakers and licence entered at project creation (FR01) must not be asked again when configuring clips (FR09) or exporting (FR19).
3.3.8 Accessible Authentication (Minimum)	AA	No authentication step requires a cognitive function test, such as remembering a password, unless an alternative is provided.	Account creation and login use email and password. The password field must allow paste and password managers. If a CAPTCHA or one-time code is added against abuse, an alternative without a cognitive test must exist.
4.1.3 Status Messages	AA	Status messages can be programmatically determined through role or properties such that they can be announced by assistive technologies without receiving focus	Asynchronous job progress updates (US06), background ingestion errors (US07) and form save confirmations must use alerts and messages, so screen readers announce them without stealing focus from the user's current task (Explained in section 2.2.4 of the report)

2.5. Video Captions and Media Accessibility

SynopStudio must be accessible at two levels: the media inside the application (players, timeline, previews) and the clips it exports to social media. Caption and effect decisions taken

in SynopStudio are inherited by every published clip, so they reach an audience far beyond our users. This section covers captions, audio and video alternatives, and animated effects.

2.5.1. Screens with media

Screen	Use case	Media elements
New Project	UC01, UC09	Spoken language, default subtitles on/off, default caption style
Video transcription	UC05, UC07	Player, synchronized transcript, speaker and topic segments
Clip proposals	UC10, UC11	Player, generated clips with source intervals, waveform timeline
Clip editor	UC12, UC13, UC14, UC17	Preview with subtitle sample, timeline (video, audio, overlay and text tracks), subtitles tab for per-clip style and translated subtitle tracks, format menu for reframing
Export	UC18	Burned-in captions, separate subtitle file per language, export presets
Viewer page	UC25	Public clip player, AI label, source interval

2.5.2. Captions and Media Alternatives

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
1.2.2 Captions (Prerecorded)	A	Recorded speech in video has captions	Every preview (transcription, proposals, editor, viewer page) must show captions built from the transcript, with the speaker's name. Since they come from the word-level transcript (FR04) and must match it verbatim (FR15), the deaf creator in 2.1.2 can approve a clip without sound.
1.2.4 Captions (Live)	AA	Live speech has captions	Not applicable: SynopStudio only processes recorded videos; streams are imported after they end.
1.2.3 / 1.2.5 Audio Description	A / AA	Visual information is available to	Clips whose visual content is not spoken (slides, on-screen text) do

		people who cannot see it	not yet meet criteria 1.2.3 and 1.2.5. A descriptive transcript (1.2.3) and later an audio description (1.2.5) are planned.
1.1.1 Non-text Content	A	Non-text content has a text alternative	The audio waveform and the clip thumbnails need a text label, such as the segment's start and end time. (Explained in section 2.2.3 of the report)
1.4.2 Audio Control	A	Audio that plays automatically can be stopped	The clip proposals screen shows several previews; must not plays sound automatically.
1.2.1 Audio-only and Video-only (Prerecorded)	A	Prerecorded audio-only and video-only content has a text alternative	The editor lets users import audio files and add a voice-over track. An audio file previewed on its own in the media library needs a text alternative, such as a transcript. Once a voice-over is added to a clip, its speech must also appear in the captions, which are otherwise built only from the original recording (1.2.2). A clip exported with its audio muted becomes video-only and needs a text description.

2.5.3. Produced Clips and Export

WCAG 2.2 Criteria	Level	Requirement	SynopStudio
2.3.1 Three Flashes or Below Threshold	A	Nothing flashes more than three times per second	Zoom cuts, word highlighting and transitions (FR17) are limited to stay below this threshold.
2.2.2 Pause, Stop, Hide	A	Moving content can be paused	Animated captions and progress bars in the editor preview stop when the player is paused.
1.2.2 Captions (Prerecorded)	A	Recorded speech has captions	Captions are on by default in every export. A separate subtitle file (FR19) is also planned, because burned-in captions are part of the image. A screen reader cannot read them, and viewers cannot turn them off or resize them.

2.6. Accessibility Matrix

The following table lists all the issues identified. Each row specifies the affected screen and element, user profile, the WCAG success criterion violated, the severity, the proposed correction.

Severity measures how much an issue affects the completion of a task by the user profile concerned and is independent of the WCAG conformance level (A or AA) of the criteria. High means the user cannot complete the task, cannot tell what state the interface is in, or ends up with a legal or irreversible consequence without being able to check it. Medium means the task can be completed but only with significant extra effort or with a real risk of error. Low means the issue causes inconvenience but has an easy workaround. When an issue affects more than one profile, the severity reflects the profile for which the impact is greatest.

Screen · Item	User profile	Problem	WCAG 2.2 Criteria	Severity	Correction
Log in · email and password fields	Low vision	Input field borders are very light grey on white (estimated ~1.5:1), making the field boundaries hard to perceive	1.4.11 Non-text Contrast	Medium	Darken field borders to a contrast of at least 3:1 against the background (e.g. #858585 or darker)
Log in · decorative waveform panel	Blind users	The waveform illustration is purely decorative but may be exposed to screen readers as a series of graphic elements, adding noise before the form	1.1.1 Non-text Content	Low	Mark the illustration as decorative: aria-hidden="true" on the SVG, or alt="" if it is an image
Log in · split-screen layout	Low vision	At 200% zoom, the decorative left panel keeps half of	1.4.10 Reflow	Medium	At narrow widths or high zoom, hide or stack the decorative panel so

		the screen, which may squeeze the form or force horizontal scrolling			the form uses the full width
Project list · Project cards	Blind users	The whole card is clickable but has no link role or name, so the screen reader reads it as loose text and the user cannot tell how to open a project	4.1.2 Name, Role, Value; 2.4.4 Link Purpose (In Context)	High	Make each card one link <a> whose accessible name is the project title, with the title as an <h2>
Project list · Search field	Blind users	The field has only the placeholder “Search…” and a magnifier icon, so it has no reliable accessible name	4.1.2 Name, Role, Value	High	Add aria-label="Search projects" (or a visually hidden <label>) and use type="search"
Project list · Filters and project count	Blind users	The selected filter is shown only by its white background, and after filtering or searching the new count (“6 projects in your library”) is not announced	4.1.2 Name, Role, Value; 4.1.3 Status Messages	High	Use buttons with aria-pressed="true" on the active filter; put the count in a live region so the new number is announced
Clip proposals · Player (captions)	Deafness	The player shows no captions and has no captions button. A deaf creator cannot check	1.2.2 Captions (Prerecorded)	High	Show captions generated from the transcript, on by default, with a button to turn them on and off

		what is said before approving the clip			
Clip proposals · Player (transcript)	Deafness, Blind users	There is no synchronized transcript next to the player, so the recording has no text alternative and speakers are not identified	1.2.3 Audio Description or Media Alternative; 1.2.2 Captions (Prerecorded)	High	Add a transcript panel with speaker names, synchronized with the player and highlighting the selected clip's interval
Clip proposals · Player (reframing area)	Blind users, Color Blindness	The 4:5 crop is shown only by an orange outline drawn over the video, with no text	1.1.1 Non-text Content; 1.4.1 Use of Color	Medium	Text such as "4:5 frame, centred on the speaker" and an outline that also differs in thickness or pattern
Clip proposals · Waveform Timeline	Blind users	The position of each clip in the original recording is shown only graphically on the waveform	1.1.1 Non-text Content	Low	The clip list already gives the intervals as text (5:00 – 5:40); link each segment to its clip and hide the decorative bars
New Project · Video Upload	Blind users; Cognitive/ low digital literacy	The state after selecting a file is not designed. Large uploads take time and progress; completion or failure is not announced	4.1.3 Status Messages	Medium	Progress bar with role=”progressbar”, status messages in a live region, and a focusable “Cancel upload” button.

		without moving focus			
New Project · Format, Subtitle style, Language, Upload/YouTube switch	Motor impairments	The selected state is an orange border with tint, and the focus ring is the same orange, so focus cannot be told apart from selection and disappears on the selected card	2.4.7 Focus Visible	High	Give focus a distinct ring (2 px dark outline with offset, never used for selection). Build each group as native radio inputs in a fieldset/legend: one Tab stop per group, arrow keys change the option.
New Project · Title, Video, Link, Supporting materials	Cognitive/ low digital literacy; Blind users	Required fields are not marked, the video size limit is missing, the listed formats (MP4, MOV, WebM) differ from FR02, and the link format is only in a placeholder	3.3.2 Labels or Instructions	Medium	Write “(required)” in text, state formats and size limit in the hint and link hints with aria-describedby instead of relying on placeholders.
Clip editor · Snapping toggle (magnet)	Color Blindness, Blind Users	On/off state shown only by the icon turning orange; no label, tooltip or state in code	1.4.1 Use of Color; 4.1.2 Name, Role, Value	High	Filled background when on, tooltip "Snapping on/off" and aria-pressed
Clip editor · Icon-only buttons (toolbar, eye, speaker, zoom, magnet)	Blind Users	Buttons show only an icon, so a screen reader has no name to announce	4.1.2 Name, Role, Value	High	aria-label on each button ("Split clip", "Hide track", "Zoom in") and a matching tooltip
Clip editor · Clips on the timeline	Motor impairments	Clips are moved and trimmed only by dragging ("Drag clips to move them and their	2.5.7 Dragging Movements; 2.1.1 Keyboard	High	Arrow keys move the selected clip; start and end times can be typed in the right panel

		edges to trim")			
Clip editor · Split, duplicate, delete, undo, redo, Save	Low vision, cognitive	Icons at 1.9:1 and "Save" at 3.8:1 look disabled with no reason given; if they are enabled, they fail	1.4.11 Non-text Contrast (AA); 1.4.3 Contrast (Minimum) (AA)	Medium	Mark as disabled in code (disabled / aria-disabled) with a tooltip such as "Select a clip first"; enabled icons use #8A8A8A (5.5:1)
Clip editor · Shortcut "S" (Split at playhead)	Motor impairments (voice control)	A single-letter shortcut can be triggered by accident, e.g. by speech input	2.1.4 Character Key Shortcuts (A)	Medium	Shortcut works only when the timeline has focus, and can be turned off or changed
Clip editor · Muted text (section labels, ruler times, track hints, "Drag clips…")	Low vision	Small (11–13 px), partly uppercase monospace text in #7C7C7C at 4.52:1, with no margin; 4.0:1 if placed on a panel	1.4.3 Contrast (Minimum) (AA)	Low	Muted text token #8A8A8A (5.5:1 on the background, 4.9:1 on panels)
Clip editor · Side panels	Low vision	The Shortcuts panel already scrolls inside its own box at 100%; at 200% zoom the panels leave little room for the preview	1.4.10 Reflow (AA)	Low	Side panels collapse into tabs at narrow widths; the timeline is exempt (two-dimensional layout)

3. References

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https://www.opus.pro/terms-of-service

https://digital-strategy.ec.europa.eu/pt/policies/regulatory-framework-ai

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https://www.w3.org/TR/WCAG22/

Sector-Specific Links

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